

Multiple Choice (10 marks)

1. What is the mole fraction of H_2SO_4 in a 7.0 molar solution of H_2SO_4 which has a density of 1.39 g/ml?
 - (a) 0.3
 - (b) 0.7
 - (c) 0.15
 - (d) 0.9
 - (e) 0.5
2. A certain radioactive material has a half-life of 36 minutes. Starting with 10.00 grams of this material, how many grams will be left after 2 hours?
 - (a) $1.5 \times 10^{-4}\text{grams}$
 - (a) 2.50 grams
 - (b) 3.33 grams
 - (c) 4.00 grams
 - (d) 1.00 grams
3. The rate law may be written using stoichiometric coefficients for which of the following?
 - (a) Redox reactions
 - (b) Elementary processes
 - (c) Dissolution reactions
 - (d) Polymerization reactions
 - (e) Complexation reactions
4. Which of the following is expected to have two or more resonance structures?
 - (a) CH_4
 - (b) NH_3
 - (c) CCl_2F_2
 - (d) H_2O
 - (e) SF_6
5. Calculate the uncertainty ΔL_z for the hydrogen-atom stationary state: $2p_z$.
 - (a) 1

- (b) 0.75
- (c) 2
- (d) 4
- (e) 0.5

True / False (10 marks)

Answer True or False

6. True or False: The molarity of sodium ions (Na^+) in human blood serum is approximately 0.15 M based on the following data.
6. True or False: The half-life of ^{32}P is 25.3 days, as determined from its decay over a specified time period.
6. True or False: Upon heating, 0.96 g of oxygen is produced from 4.5 g of potassium chlorate (KClO_3), consistent with stoichiometric calculations.
7. True or False: Lead is a liquid element that appears dark-colored and is nonconducting at room temperature.
8. True or False: The pH of a 0.0001 N HCl solution is 10, suggesting it is a neutral solution.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Given an unknown element X with an atomic weight of 210.197 amu and isotopes ^{210}X (mass 209.64 amu) and ^{211}X (mass 211.66 amu), calculate and discuss the relative abundances of the isotopes.
11. In a series of two electrolytic cells containing AgNO_3 and CuSO_4 , analyze the electrochemical processes occurring when 1.273 g of Ag is deposited. Calculate the mass of copper deposited concurrently and discuss the principles involved.

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